

HOMEWORK 10

Problem 1

Consider the nonlinear regression model,

$$y_i = m(x_i) + \sigma\epsilon_i, \quad i = 1, 2, \dots, n,$$

where $n = 100$ and the data (x_i, y_i) are provided in “HW10_data.csv”.

- (1) (10 pts) Using the uniform kernel giving weights of the form

$$w_{i,h}(x) = \frac{\mathbf{1}(|x_i - x| < h)}{\sum_{i=1}^n \mathbf{1}(|x_i - x| < h)},$$

find the estimator of $m(0)$; i.e. $\hat{m}(0)$, obtained by minimizing

$$\sum_{i=1}^n w_{i,h}(0) (y_i - \theta)^2$$

using $h = 0.5$, with respect to θ .

- (2) (10 pts) Write down an expression for and hence find the value of the bias of $\hat{m}(0)$ if the true function $m(x)$ is linear with slope 1.
- (3) (10 pts) What is the variance of $\hat{m}(0)$? Leave your answer in terms of the unknown σ^2 .
- (4) (10 pts) Compute the estimator for the variance, given by

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \hat{m}(x_i))^2$$

- (5) (10 pts) Using the value of $\hat{\sigma}^2$, find, using a normal approximation, the 95% confidence interval for the predictive value of y at $x = 0$.

Problem 2

Consider the nonlinear regression model, with $n = 100$ and data (x_i, y_i) provided in “HW10_data.csv” (the same dataset as Problem 1),

$$y_i = m(x_i) + \sigma\epsilon_i, \quad x_i \in (-3, 3), \quad i = 1, \dots, n,$$

and the aim is to split the range of $x \in (-3, 3)$ into two intervals; the start of a regression tree. That is, we are looking for the estimator

$$\hat{m}(x) = \begin{cases} m_1 & x \leq r \\ m_2 & x > r. \end{cases}$$

- (1) (10 pts) Find the optimal choice of r which minimizes the sum of square errors. Show your working and algorithm.
- (2) (10 pts) What is your result from (1)? State your optimal choice of r and $\hat{m}(x)$ function.
- (3) (5 pts) What is the predictive value for the y corresponding to $x = 0.5$?
- (4) (15 pts) Using the non-parametric bootstrap, find an estimator for the variance of $\hat{m}(0.5)$. Provide all your working.
- (5) (10 pts) Using the answer to part (4), construct an approximate 95% confidence interval for $m(0.5)$.